Search for the ^{4}He $-\eta$ bound state with WASA-at-COSY

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Abstract. We propose to search for the ^{4}He $-\eta$ bound state via measurement of the excitation functions for the $dd \to {}^3\mathrm{H}p\pi^0, {}^3\mathrm{H}n\pi^+, {}^3\mathrm{He}n\pi^0, {}^3\mathrm{He}p\pi^-$ reactions where the outgoing $N-\pi$ pairs originate from the conversion of the η meson on a nucleon inside the ^{4}He nucleus. Precise determination of the profile of the expected Breit-Wigner distribution in the excitation curves will allow to determine the binding energy and the width of the ^{4}He $-\eta$ bound state.

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The theoretical predictions and the threshold enhancements in reactions leading to ^{3}He $-\eta$ and ^{4}He $-\eta$ final states show that the low-energy interaction between the η meson and the He nucleus is attractive and strong [1, 2, 3]. Those facts can be seen as an indication of the possible existence of meson-nucleus bound states. Nevertheless, up to now there is no clear experimental evidence for such states. First experimental indications of the ligth η-nucleus bound state was reported at MAMI in the η photoproduction reaction $\gamma^3\mathrm{He} \to \pi^0 pX$ [4]. Difference between excitation functions for two ranges of $\pi^0 - p$ relative angle in the center of mass (c.m.) frame showed structure which was interpreted as a possible signature of the bound state. However, it was claimed that due to the limited statistics the result could be interpreted as an evidence of virtual state [5]. Another indication of the existense of the quasi-bound state of the ^{3}He $-\eta$ system was recently discussed by Wilkin [6, 7] and collaborators based on the excitation function and on angular asymmetries of the $dp \to {}^3\mathrm{He} - \eta$ reaction measured recently at COSY [8, 9]. But the clear empirical confirmation is still missing.

We propose to extend the research to study the possible ^{4}He $-\eta$ bound state produced in the $d-d$ collisions close to the kinematical threshold for the $dd \to {}^4\mathrm{He}\eta$ process with the WASA-at-COSY detector [10]. According to the calculations including multiple scattering theory [11] and the Skyrme model [12], the ^{4}He $-\eta$ system is especially good candidate for the bounding. It is expected that the ^{4}He $-\eta$ system will decay via excitation of the S11(1535) resonance in absorption of the η meson on one of nucleons in the ^{4}He nucleus. The resonance will subsequently decay in $N-\pi$ pair. We propose to study the metioned decays in one of the four channels:

- $d+d \to ({}^4\mathrm{He}-\eta)_{bound} \to {}^3\mathrm{H}p\pi^0$,
- $d+d \to ({}^4\mathrm{He}-\eta)_{bound} \to {}^3\mathrm{H}n\pi^+$,
- $d+d \to ({}^4\mathrm{He}-\eta)_{bound} \to {}^3\mathrm{He}n\pi^0$,
- $d+d \to ({}^4\mathrm{He}-\eta)_{bound} \to {}^3\mathrm{He}p\pi^-$.

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For the first and fourth channel an inclusive measurement is possible, whereas for the second and third one kinematically complete measurement can be performed by the detection of spectator nucleus and one of the other outgoing particles and by using the missing mass technique. The emission angle between the nucleon and pion in the $d-d$ c.m. system is around 180 degrees and is smeared by a few degrees due to the Fermi motion of the nucleon inside the ^{4}He nucleus. The c.m. kinetic energies of the nucleon and pion orginate from the mass deficit $m_\eta - m_\pi$ and are around 50 MeV and 350 MeV, respectively. In turns, the ^{3}He/^{3}H ejectile is a spectator moving with Fermi momentum in the c.m. frame. The smearing of the spectator momenta is around 50 MeV/c which is much smaller than the mean momentum in the laboratory frame (around 1500 MeV/c). Therefore, in the laboratoty system the ^{3}He/^{3}H ejectiles are confined in a cone opened in the forward direction with an angle of around 2 degrees.

For estimation of the counting rate in the proposed measurements we make an assumption that the maximum cross section for $dd \to (^4\mathrm{He}-\eta)_{bound}$ production is about 150 nb. This value is about ten times larger than the $dd \to {}^4\mathrm{He}\eta$ cross section near threshold ($\sim$ 15 nb) which is in line with the observed ratio of the $\gamma^3\mathrm{He} \to (^3\mathrm{He}-\eta)_{bound} \to p\pi^0 X$ and $\gamma^3\mathrm{He} \to {}^3\mathrm{He}\eta$ cross sections measured at MAMI [4]. We assume that the probability of decay of the $^4\mathrm{He}-\eta$ bound state into one decay channel is equal $\frac{1}{4}\cdot\frac{1}{2}=\frac{1}{8}$. The factor $\frac{1}{4}$ takes into account the fact that we measure only one out of four possible η absorption channels. In turns, factor $\frac{1}{2}$ represents our guess of the probability that the three spectator nucleons form ^{3}He/^{3}H in the final state. We justify our guess using an analogy between the decay of the $^4\mathrm{He}-\eta$ nuclei and $^4_\Lambda$He - hipernuclei. For the later case it was observed namely that in the π^- decay channel the decay mode $^4_\Lambda\mathrm{He} \to \pi^- p\,{}^3\mathrm{He}$ is dominant [13]. With a luminosity of $10^{31}\mathrm{cm}^{-2}\mathrm{s}^{-1}$ and with the WASA-at-COSY acceptance for detection of the $p-\pi^0$ pairs close to 100%, the number of registered $d+d \to (^4\mathrm{He}-\eta)_{bound} \to {}^3\mathrm{H}p\pi^0$ events will be 16 200 per day.

The experiment would be performed with the internal deuteron beam of COSY scattered on the WASA-at-COSY deuteron target of the pellet type. We propose to use slowely ramped deuteron beam of momentum in the range 2.219 GeV/c to 2.36 GeV/c to scan the cross sections. It corresponds to excess energy Q in the $d+d \to (^4\mathrm{He}-\eta)$ reaction from -40 MeV to 10 MeV and this range is about two times wider than the width of the structure observed at MAMI in the $\gamma^3\mathrm{He} \to \pi^0 pX$ process (25.6 ± 6.1 MeV). A signature of existence of the $^4\mathrm{He}-\eta$ bound state will be based on the observation of a resonance like structure in the excitation curves for chosen decay channels cross sections. A fit of the Breit-Wigner curve to the experimental data should deliver the binding energy and the width of the bound state.

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